

REPOSITORY HARDENING REPORT

512 / CVS (Evidence-Sidecar) / Constraint-Architecture

Date	24 March 2026
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Pass type	Architectural hardening, anti-drift, licensing, developer navigation
Repos affected	3 — 512, Evidence-Sidecar (CVS), Constraint-Architecture
Total files changed	19 files across three repositories
Archive hashes	Three independent SHA-256 seals recorded

1. Executive Summary

This report documents the hardening pass conducted on 24 March 2026 across three repositories: 512, Evidence-Sidecar (CVS), and Constraint-Architecture. The pass addressed architectural drift vectors, language violations, invariant count discrepancies, licensing gaps, developer navigation, and upstream referencing. The work was driven by real-world feedback from external builders and enterprise contacts who correctly identified every place where the architecture could be bent or reinterpreted. The hardening closes those surfaces.

All three repositories are now sealed with independent SHA-256 archive hashes recorded on 2026-03-24.

2. What Changed — by Repository

2.1 CVS (Evidence-Sidecar)

9 files changed — 1 modified, 8 new.

File	Action	What it addresses
CONFORMANCE.md	Modified	Added operational states (Verified/Degraded/Halted), binary test checklist (14 tests), non-goals section, constraint formation responsibility
ANTI_DRIFT.md	New	15 sections — layer separation, fixed observation model, no interpretation at capture, evidence formation constraint, witness isolation, no control loop,

		sufficiency, multi-party verification, language reference
VERIFICATION_PROTOCOL.md	New	6-step canonical verification protocol for auditors, regulators, and independent reviewers
02_EVIDENCE_MODEL/PROOF_OBJECT_INTEGRITY.md	New	4 required elements, self-containment requirement, schema position — no canonical schema
02_EVIDENCE_MODEL/SPEC_HASH_BINDING.md	New	Binding lifecycle — initialisation, per-object embedding, divergence handling (3 options)
03_SELECTIVE_DISCLOSURE/DISCLOSURE_KERNEL_CLARIFICATION.md	New	Proof minimisation vs access control — explicit prohibition of RBAC/IAM implementations
LEGAL_NOTICE.md	Replaced	Apache 2.0 attribution, three-tier open commons declaration, prohibited terms removed
README.md	Modified	Hardening banners removed, CVS-Conforming language, normative docs listed
UPSTREAM.md	New	Three-layer stack reference, Constraint-Architecture pointer

2.2 512

9 files changed — 4 modified, 5 new.

File	Action	What it addresses
README.md	Modified	START HERE developer navigation block, kernel pointer corrected, properties language, banners removed
512-core/KERNEL/INVARIANTS.md	Modified	Corrected to 7 invariants — Data Sovereignty removed (not in kernel text), Binary Adherence merged into I7

512-core/CANON/KERNEL_EQUIVALENCE_AND_SPEC_HASH.md	Modified	Corrected from I1-I9 to I1-I7, normalization rules for merged invariants added
512-core/CANON/README.md	Modified	Adoption: Voluntary removed, I1..I9 corrected to I1..I7, properties language
512-ops/COMPLIANCE_CHECKLIST.md	Modified	Renamed to Properties Checklist — compliance framing removed
ANTI_DRIFT.md	New	12 hardening additions — temporal constraint, policy engine kill, constraint definition separation, observation mode, no advisory, intent-execution correspondence, single authority path, determinism, gap semantics, binary satisfaction
512-ops/COMMIT_BOUNDARY_REFERENCE.md	New	Developer worksheet — where is the boundary, Proposal Object, 7-invariant evaluation, 3 outputs, XRPL anchoring
LICENSE	New	Root-level CC BY 4.0 for documentation, explicit constraint set non-ownership declaration
UPSTREAM.md	New	Three-layer stack reference, Constraint-Architecture pointer

2.3 Constraint-Architecture

1 file changed.

File	Action	What it addresses
README.md	Modified	CVS name corrected (System → Sidecar), 512-Compliant Consultant → 512 Integration Specialist, three-layer stack table with canonical repo links, repository structure section completed

3. Drift Vectors Closed

These are the specific attack surfaces that were identified and hardened. Each represents a place where a vendor, enterprise, or regulator could have reinterpreted the architecture.

Drift Vector	How it manifests	Where hardened
Configurable observation surface	Vendors inserting custom capture layers and claiming compatibility	ANTI_DRIFT.md §2 — fixed observation model, three event types only
Schema hijacking	Third party publishing a fixed CVS schema and claiming it as canonical	PROOF_OBJECT_INTEGRITY.md — invariant-defined minimum, no canonical schema
Disclosure as access control	Enterprise teams implementing Disclosure Kernel as RBAC/IAM	DISCLOSURE_KERNEL_CLARIFICATION.md — proof not permissions
Silent degraded execution	Systems continuing without evidence and not declaring the gap	CONFORMANCE.md — operational states, silent execution is non-conformant
Upstream constraint layer invisible	Vendors quietly taking ownership of constraint definition	ANTI_DRIFT.md §5, UPSTREAM.md — constraint definition is external discipline
No interpretation at capture	Light interpretation embedded at CVS capture layer	ANTI_DRIFT.md §6 — capture layer must not interpret, classify, or derive
Evidence reconstruction	Logs assembled post-hoc and described as Proof Objects	ANTI_DRIFT.md §7 — evidence formed at execution time only
Witness isolation	CVS participating in constraint definition	ANTI_DRIFT.md §8 — witness must not define or evaluate constraints
Control loop from witness	CVS feeding signals back into execution path	ANTI_DRIFT.md §9 — CVS is passive, no feedback
Policy engine substitution	512 replaced with dynamic policy interpretation	512/ANTI_DRIFT.md — deterministic only, no scoring or weighting
Advisory mode	Boundary returning recommendations instead of allow/deny/gap	512/ANTI_DRIFT.md — three outputs only, no advisory
Bypass accumulation	Admin paths and overrides accumulating over time	512/ANTI_DRIFT.md — single authority path, no parallel execution

4. Invariant Count Resolution

A significant discrepancy was found and resolved during this pass. Three files in the 512 repo referenced different invariant counts:

Document	Count	Status
512-kernel.txt.txt (canonical)	7 statements	The authority — unchanged
INVARIANTS.md (before)	8 invariants	Split I7 into two — fixed
KERNEL_EQUIVALENCE (before)	9 invariants (I1-I9)	Further split I3 and I7 — fixed
STYLE_GUIDE	7 Lockean invariants	Confirmed correct

Resolution: all documents now reference 7 invariants (I1-I7), mapped directly to the 7 kernel statements. Data Sovereignty was removed from INVARIANTS.md — it was not present in the kernel text and its inclusion violated FORBIDDEN_MUTATIONS.md §II.

5. Licensing and Ownership Clarification

Licensing was inconsistent across the repos before this pass. The following is now clear and consistent:

What	Status	Implication
512 constraint set	Not owned. Not ownable. Discovered, not invented.	No licence required to build against it
512 documentation	CC BY 4.0 — Jonathan M. Watson	Attribution required for reproduction
CVS base architecture	Apache 2.0 — Jonathan M. Watson	Patent grant, attribution required, commercial use permitted
CVS derivatives	Owned by derivative authors	Freely commercialisable under Apache 2.0 terms

Root LICENSE file added to 512 repo. LEGAL_NOTICE.md replaced in CVS repo with full Apache 2.0 attribution and open commons declaration.

6. The Three-Layer Stack

This pass formalised the three-layer stack and established explicit upstream references between all three repos:

Layer	Function	Repository
Constraint Architecture	Defines what is admissible — consent logic, authority models, thresholds, domain constraints	github.com/JonathanMastersWatson/Constraint-Architecture

512	Enforces admissibility at the commit boundary — binary, deterministic, at machine speed	github.com/JonathanMastersWatson/512
CVS	Witnesses execution and produces independently verifiable evidence	github.com/JonathanMastersWatson/Evidence-Sidecar

UPSTREAM.md added to both 512 and CVS repos. Constraint-Architecture README updated with canonical stack table and repo links. Key principle added to all three repos:

"Governance is not achieved by observing behaviour. It is achieved by constructing the boundaries within which behaviour is possible."

7. Archive Hashes — 2026-03-24

All three repositories were downloaded as fresh zips following the hardening pass and independently hashed. These hashes record the exact state of each repo at seal.

Repo	File	SHA-256
512	512-main-2026-03-24.zip	DF27F6C5C8DDBBD5341FB15EA943D92B3388331B386C26F44A145673F6C8D218
CVS	Evidence-Sidecar-main-2026-03-24.zip	0DA1ABA3A7257B636C0A364920C5CE643843257D217C0DBD68BA5DB64712BE17
Constraint-Architecture	Constraint-Architecture-main-2026-03-24.zip	2DA95899744AFCFF3A9C13AADFB47BE11166BDC262002A53CA7E8B46964F6CE5

Verification method (PowerShell):

```
(Get-FileHash "<filename>.zip" -Algorithm SHA256).Hash
```

Hash records committed to each repo in dedicated proof files.

8. For Builders — What This Means

If you are building against 512 or CVS, the following files are your primary references after this hardening pass:

512 — Start Here

- 512-core/KERNEL/512-kernel.txt.txt — the 7 invariant statements (the authority)
- 512-core/KERNEL/INVARIANTS.md — each invariant defined precisely
- 512-ops/COMMIT_BOUNDARY_REFERENCE.md — where is the boundary, what crosses it, what comes out
- ANTI_DRIFT.md — what 512 is not, how implementations drift

- UPSTREAM.md — the three-layer stack and your responsibilities upstream

CVS — Start Here

- CONFORMANCE.md — what CVS-Conforming means, binary test checklist
- ANTI_DRIFT.md — layer separation, fixed observation model, prohibited patterns
- VERIFICATION_PROTOCOL.md — how to verify evidence as a third party
- 02_EVIDENCE_MODEL/PROOF_OBJECT_INTEGRITY.md — what a Proof Object must contain
- UPSTREAM.md — CVS position in the stack, constraint definition is upstream

Key Principle for All Implementations

Errors in constraint definition are upstream failures. 512 will faithfully enforce incorrectly defined constraints. CVS will faithfully record adherence to incorrectly defined constraints. That is by design. Define your constraint boundary correctly upstream before touching either system.

9. What Remains

The following items were noted during this pass and are deferred to a future pass:

- 512-ops/REFERENCE_FLOW.md — still a placeholder, low priority
 - Legal review package (COVER_LETTER.md, EXECUTIVE_BRIEF.md, LEGAL_QUESTIONS.md, TENSIONS_SUMMARY.md) — to be built as private briefing for legal counsel
 - constraint-architecture-foundations repo — exploratory foundations repo, separate from canonical repos, needs its own pass
 - Layer 2 product surface documents (Vyacheslav's control plane proposals) — derivative work, not canonical, separate repo recommended
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10. Closing Statement

The architecture is now significantly harder to distort. The hardening pass removed every place where a vendor could insert themselves into a layer they do not belong, every place where an enterprise could reinterpret the architecture, and every place where a regulator could map it to something it is not.

The three canonical repos — 512, CVS, and Constraint-Architecture — now form a coherent, cross-referenced stack with consistent terminology, correct invariant counts, clean licensing, and explicit anti-drift documentation.

Systems that build on this architecture now have a precise, verifiable target. That was the goal.

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